

Personalised AI Assisted Masters Programme Selection

Business and Technical Trade off Analysis | NIU DUOER | PE6201 Section B | 4 October 2026

Business and Technical Trade-off Analysis

The problem I chose

DS Compass is for an international graduate who has a broad subject in mind and wants a manageable first list of master's programmes in Singapore and the United Kingdom. The difficult part is translating a goal such as "AI" or "business analytics" into programme names, then checking each page for entry conditions and English scores.

My project narrows that first step. A user chooses a direction and country, and the app retrieves matching records from a small catalogue with links to university pages. It then puts the recorded academic requirement and English evidence beside each result. It does not rank universities or estimate admission chances. I reviewed the supplied historical examples, which include some UK and Singapore offers, but they mix countries, universities and broad programme labels and are largely positive outcomes. The available sample does not provide enough comparable offers and unsuccessful outcomes to calibrate a fair tier from university background and GPA. I decided that a small, auditable official-source workflow was a stronger first release than a weak Reach/Match/Safer estimate. I then expanded the early Singapore version to include the UK, as the proposal's cross-country goal required.

Decisions made during development

The proposal described a foundation model working with official-source retrieval. In the final system, the model has a much narrower job: it classifies short text only when the user chooses "Other or unsure." It returns one of six fixed directions or 'abstain'. A chosen direction bypasses the model and filters 'data/programmes.json' directly. The programme facts always come from that local file, not from a generated answer. I chose this division after considering the risk of a fluent model inventing an English score or course requirement. A fixed label gives the model less room to make that kind of mistake.

The first form also changed after review. The institution list was too short and skewed toward famous universities, and the results looked repetitive because there were few programme records. I expanded the school suggestions and the catalogue, though 15 records are still not broad coverage. I removed the IELTS test-date and planned course-start fields, separated the four IELTS components, and added a TOEFL option. The school and 985/211/Double First Class fields are visible, but they do not influence retrieval. I could not support a fair ranking from the available evidence, so I left them as context.

One bug was particularly easy to miss: selecting a fixed direction such as AI initially still triggered the goal-text validation, because the server expected a longer text goal. I changed the request handling so fixed directions map to an internal label before validation. This let the form stay simple without sending misleading free text. Another rule change was to return 'review' when an IELTS threshold is missing from a catalogue record. A blank field cannot safely mean that an applicant has met the requirement.

IELTS overall and component scores are compared only where a source supports an encoded rule. TOEFL is stored but not converted to IELTS. I also added a simple profile-to-requirement layer: the selected major is labelled as broadly related to the recorded academic requirement, potentially suitable for a conversion route, or requiring manual review. The requirement itself is visible on the card. I checked this rule against eight hand-specified profiles across AI, data, business, education and conversion courses; all eight returned the expected evidence label. That is a regression check for the rule, not validation of admissions. Academic equivalence, transcript detail, exemptions and current test validity remain with the university. The evidence labels are therefore not Reach, Match or Safer predictions.

What the data and evaluation show

The programme file contains 15 short paraphrases with an official URL and the date checked. This makes it possible to follow a card back to its source, but it does not make the catalogue complete. Some directions have only one or

two results, and a university can change a page at any time. The 90-day warning is a reminder to recheck, not a data-refresh system. A second reviewer should check the encoded rules before this is used beyond a class prototype.

I evaluated the keyword classifier on 37 synthetic prompts. GPT-4.1 mini helped write the prompts and their reference labels before I ran Gemini 2.5 Flash Lite. The returned set had 37 valid cases from 42 requested; I kept that set rather than adding examples after seeing the results. I had not set a minimum accuracy target in advance, so the score is descriptive. Gemini matched 32 labels, or 86.5%; an always-`abstain` baseline matched 7 of 37, or 18.9%. The 37 model calls cost about USD 0.000542 according to the provider's usage record. That is the cost of this run only, not a promise about future pricing.

The accuracy number leaves out the failure I care about most. Seven prompts were labelled out of scope; Gemini abstained on only three and assigned an in-scope direction to the other four. In T11, a question about improving a company's sales pipeline was expected to be `business` but returned `data`. These errors show that a fixed output schema does not by itself make boundary detection reliable.

The test set is synthetic, small and model-assisted. It is not an independent human benchmark, and its wording may resemble the model's training patterns. The baseline is deliberately weak, so beating it is only a basic check. I did not run a separate calibration study or tune a threshold against these cases; the local fallback's cut-offs remain heuristics. The current results support a narrow conclusion: Gemini classified many of these examples correctly, while out-of-scope recall was poor at three of seven. They say nothing about admission outcomes or whether an applicant found a better programme.

Trade-offs and next work

For a model failure, the worker uses a local Laplace-smoothed Naive Bayes classifier and shows that it is the fallback. It is inexpensive and easy to rerun, but the tokeniser recognises English words only; it is not a Chinese-language alternative. OpenRouter keeps the hosted integration straightforward, while introducing an external dependency, possible outages and a secret that must stay server-side. The worker constrains the model response to fixed labels and limits request size and rate.

The next useful step is a small set of consented applicant queries with human labels and an untouched test split. A second reviewer should also check programme and English-rule entries. Until then, DS Compass is a source-linked starting list, not a tool to trust without review.

Sources

The programme URLs and language-rule URLs are stored with each record in `data/programmes.json`. The evaluation prompts, predictions and calculation notes are in `evals/`.